6.2.4 Apache SkyWalking

Apache SkyWalking (https://skywalking.apache.org/) is an ASF open source project that, nowadays, offers support for distributed tracing of distributed systems, like microservices in Kubernetes environments. It started out as a distributed tracing system in China Unicom in 2015 and has been adopted by over 100 companies, including Alibaba, China Merchants Bank, and Huawei. If you want to dive deeper into the topic, check out "Application Performance Monitor and Distributed Tracing with Apache SkyWalking in Datahub" (http://mng.bz/qr7r) by Liangjun Jiang to see it in action. With that, we move on to two all-in-ones that build atop of ClickHouse.

6.2.5 SigNoz

SigNoz (https://github.com/SigNoz/signoz) positions itself as an open source alternative to Datadog and New Relic. Currently, it uses ClickHouse as its backend; until mid-2021, it focused on an Apache Kafka and Apache Druid combo (http://mng.bz/7DBv) for storage. See also chapter 5 for more details on these backends. SigNoz is easy to set up and comes with several tutorials, covering topics from consuming Kubernetes infrastructure metrics (http://mng.bz/mV7a) to visualizing JVM metrics from a Spring Boot application.

Let's have a very quick walk-through to see SigNoz in action. All you need for that is to have Docker Compose installed and then execute the following:

```
$ git clone -b main https://github.com/SigNoz/signoz.git && cd signoz/deploy/
$ docker-compose -f docker/clickhouse-setup/docker-compose.yaml up
```

If you now wait a couple of minutes (to give the bundled app some time to generate signals), head over to http://localhost:3301/, and select a trace, you should see something akin to what is shown in figure 6.9 (SigNoz's trace view). Next up is another ClickHouse-based open source all-in-one offering.

6.2.6 Uptrace

Uptrace (https://github.com/uptrace/uptrace) is an all-in-one tool that states it support traces, metrics, and logs and uses OpenTelemetry to collect data as well as ClickHouse to persist the telemetry signals with the capability to offload cold data (historical data you don't frequently access) to S3-compatible object storage. For alerting, Uptrace has integrations with several notification channels, including email, Slack, and Telegram.

Uptrace has a positioning and architecture similar to what you've seen with SigNoz, and while at time of writing it seems less mature than SigNoz, there's certainly potential here. I wouldn't be surprised to see even more open source all-in-ones using a similar setup going forward. With this, we conclude the open source offerings and turn our attention to commercial offerings in the space.